

PAKO DIALE

Health Systems Science & Health Analytics Graduate | Data Analysis | Python

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PROFESSIONAL SUMMARY

Health Analytics graduate from the University of the Witwatersrand with training in Python, statistical analysis, and machine learning. Experienced in analysing health datasets, building predictive models, and identifying patterns using supervised and unsupervised learning techniques. Internship experience at CSIR supporting Health Technology Assessment and Quality Management Systems. Seeking a graduate data or health analytics role where analytical skills can support data-driven decision making.

SKILLS

Programming	Python (Pandas, NumPy, Scikit-learn), R
Machine Learning	Logistic Regression, Decision Trees, K-Means Clustering, Model Evaluation
Data Analysis	Exploratory Data Analysis, Data Cleaning, Statistical Analysis
Visualization	Matplotlib, Seaborn, Excel
Tools	Microsoft Excel, PowerPoint, Word
Health Analytics	Health Technology Assessment (HTA)
Quality Systems	ISO 13485, ISO 9001, Document Control, SHEQ Compliance

EDUCATION

Bachelor of Health Sciences (Honours) — Health Systems Science

University of the Witwatersrand, Johannesburg | 01/2025 – 12/2025

Research project: *Climate change trends and health system responses on malaria prevalence and incidence in South Africa: A Scoping Review*. Modules: Contemporary Health, Innovative Health, Research Projects, Research Methodology, Health Analytics (Python), Responsible Leadership, Systems Health.

Bachelor of Science — Animal, Plant & Environmental Sciences

University of the Witwatersrand, Johannesburg | 01/2021 – 12/2024

Final-year project: *Grazing Intensity and Landscape Function: A Comparative Study*, applying Excel and R for statistical analysis and environmental data interpretation. Majors: Biodiversity Science, Applied Ecology, Global Change, Ecosystem Dynamics.

National Senior Certificate

Thethe High School | 12/2020

HEALTH ANALYTICS PROJECTS (PYTHON)

Heart Disease Prediction Model — Logistic Regression

Health Analytics Coursework Project

- Built a supervised ML model to predict the presence of heart disease using demographic and clinical variables
- Performed data cleaning/preprocessing and exploratory analysis (correlation heatmaps, histograms, count plots)
- Applied feature scaling and an 80/20 stratified train-test split; evaluated with accuracy, precision, recall, F1-score and confusion matrix
- Achieved 80.3% prediction accuracy — implemented in Python, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn

Air Pollution Clustering Analysis — K-Means

Health Analytics Coursework Project

- Analysed an air quality dataset (PM2.5, PM10, NO2, SO2, O3) to detect pollution patterns
- Preprocessed data using StandardScaler; applied K-Means clustering
- Determined optimal clusters via the Elbow Method and Silhouette Score

- Identified three distinct pollution profiles linked to particulate matter and traffic emissions

PROFESSIONAL EXPERIENCE

Intern / Student Visitor — Quality Management Systems & Health Technology Assessment

Council for Scientific and Industrial Research (CSIR) | 07/2025 – 11/2025

- Gained practical exposure to Quality Management Systems (ISO 13485 & ISO 9001) in a regulated research and manufacturing environment
- Assisted with Health Technology Assessment documentation using evidence-based frameworks
- Supported document control processes including creation, revision, and version management
- Participated in supplier verification and ISO certification checks
- Observed internal audits, gaining exposure to compliance and continuous improvement processes

ACADEMIC & VOLUNTEER EXPERIENCE

Teaching Assistant — Health Systems

University of the Witwatersrand, Johannesburg | 07/2025 – 11/2025

- Facilitated tutorials and group discussions for first-year Health Systems students
- Supported academic writing, referencing, and research skill development
- Assisted with course administration and preparation of teaching materials

Volunteer — YEBO GOGGA Science Exhibition

University of the Witwatersrand, Johannesburg | 05/2024

- Promoted science awareness and education to learners and the public
- Supported coordination of exhibition activities and visitor engagement